Learning Energy-Harvesting Soaring Policies in Time-Varying Turbulent Wind

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Abstract

Soaring converts spatial and temporal variation in the atmosphere into usable energy, but learning such behavior is difficult when the useful signal is local, noisy, and coupled to vehicle dynamics. We introduce ThermalHunter, a controlled numerical platform for studying energy-harvesting soaring in time-varying turbulent wind. The platform couples DNS snapshots of Rayleigh–Benard convection at $Ra = 5 \times 10^7$ to two complementary flight models: a transparent quasi-steady discrete controller and a higher-fidelity dynamic aerodynamic environment. In an exploratory steady-regime comparison that reused the model-selection scenarios, a tuned DQN achieved a mean height change of 337.4 m (bootstrap 95% CI $[297.4, 376.2]$), compared with -107.1 m for random control. We also show that DQN action maps can suggest sensor thresholds for an interpretable tabular Q-learning controller; that derived controller attained 192.3 m on the same scenarios. In the higher-fidelity dynamic regime, three PPO replicas evaluated on a new 100-scenario suite had mean height changes of 397.7, 276.8, and 378.5 m, while three dynamic-DQN replicas had -146.3, -121.6, and -76.0 m. These results support energy-harvesting behavior in the simulated turbulent flow, but not route navigation, safety, real-flight deployment, or sim-to-real transfer.

1 Introduction

Unpowered gliders can gain energy by exploiting atmospheric motion. The problem is attractive for reinforcement learning (RL) because the controller must repeatedly choose how to trade local lift, turning, and aerodynamic loss without an externally supplied trajectory. Field work has shown that an RL controller can support autonomous glider soaring (Reddy et al., 2018), while simulation studies have demonstrated that learned policies can control gliding trajectories with nontrivial aerodynamics (Novati et al., 2019). Those results motivate a more controlled question: can an agent learn to harvest energy from time-varying, turbulent flow using only quantities that could plausibly be available to an onboard controller?

ThermalHunter addresses that question with a numerical experimental platform. Its wind input consists of direct numerical simulation (DNS) snapshots of Rayleigh–Benard convection at $Ra = 5 \times 10^7$. The wind is therefore simulation data, not an atmospheric measurement or a model of a particular flight test. The platform exposes local vertical-wind acceleration and a wing-span vertical-wind difference, rather than the complete flow field, and uses height change and total-energy-height change as outcomes. It is designed to study energy extraction from a turbulent environment, not to optimize a route or demonstrate a safety-critical autonomous aircraft.

The study has two complementary fidelity regimes. The first combines a quasi-steady glider model with discrete angle-of-attack and bank commands. It provides a small, inspectable testbed for comparing tabular Q-learning and DQN. The second retains dynamic flight and actuator states and

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tests whether the learned energy-harvesting behavior persists under a richer model. The regimes are not algorithmic leaderboards: policies are compared only within the environment in which they were trained and evaluated.

This manuscript makes three bounded contributions. First, it specifies a two-fidelity RL testbed driven by a single DNS wind source and matched-scenario evaluation. Second, it establishes a steady-regime baseline in which the DQN is evaluated against random and tabular-Q controllers under identical initial conditions. Third, it treats the DQN action map as a representation-design tool: its coarse decision boundaries are used to choose transparent sensor bins for a tabular controller. The last comparison is reported as preliminary, because it has not yet been replicated across independent training seeds or an untouched test suite.

2 Related work

RL has been applied to flight control and to decision-making in complex flows. Reddy et al. (Reddy et al., 2018) demonstrated glider soaring via RL in the field, linking learned control to an operational aircraft setting. Novati et al. (Novati et al., 2019) used deep RL for controlled gliding and perching, showing how learning can accommodate aerodynamic control objectives. In a different flow-physics setting, Colabrese et al. (Colabrese et al., 2017) showed that RL can discover navigation strategies from local observations in vortical flows. ThermalHunter shares the local-observation premise but changes the objective: the primary outcome is energy or altitude gain, not displacement toward a target.

The present work also uses a standard value-based deep RL baseline. DQN combines experience replay and a target network to learn action values from high-dimensional observations (Mnih et al., 2015). Here its role is deliberately modest. The DQN is not presented as a universal control solution; instead, it offers a continuous sensor-state policy against which a compact discretization can be inspected. That comparison exposes a practical experimental question: whether a learned action map can help select a lower-dimensional, auditable state representation for tabular learning.

3 ThermalHunter environments

3.1 Wind input and scaling

The shared input is a sequence of three-dimensional velocity fields from DNS Rayleigh–Benard convection at $Ra = 5 \times 10^7$. At each query, the simulator uses trilinear spatial interpolation; the dynamic environment additionally interpolates between adjacent temporal frames. Horizontal position is periodic in the $1000 \times 1000 \times 1000$ m numerical domain. To give the flow a consistent relation to the vehicle, the simulator scales the three-dimensional wind so that its RMS speed is 0.5 times the trim true airspeed at the median angle of attack. This scaling is a modeling choice, not a claim about an observed atmosphere.

Each rollout begins from a randomized horizontal position, altitude, and heading. During the steady-regime evaluation, the initial wind frame is drawn from frames 60–99, excluding the first 60 DNS frames. The initial altitude is drawn uniformly from 20–60% of the domain height. Episodes terminate at the lower or upper altitude bound (10% or 90% of the domain height) or when the wind sequence ends. We do not define a 100 m success threshold; results are reported as continuous distributions and, in the completed dynamic study, will also include termination reasons.

3.2 Quasi-steady discrete regime

The completed regime represents a glider with a quasi-steady polar. For an angle of attack α and bank angle μ, the model obtains lift and drag coefficients from the polar and solves for trim airspeed V, glide angle γ, and turn rate. A control change incurs a multiplicative drag penalty of 1.1. The controller acts every 1 s, with two physics integrations per control step. Wind data advance once every two control steps. Thus the flight model is quasi-steady, but the agent still flies through a time-varying DNS sequence.

Actions are the nine combinations of incremental changes $(\Delta\alpha, \Delta\mu) \in \{-1, 0, 1\}^2$. Angle of attack spans 0–9 degrees in 3-degree increments, and bank spans -20 to 20 degrees in 10-degree increments.

The observation contains the current control indices, the local vertical-wind acceleration a_w, and the difference in vertical wind between the two wing tips Δw. The reward is

$$r_t = w_z(t) + 5a_w(t), \tag{1}$$

where w_z is local vertical wind. This reward guides learning; it is not the primary reported outcome.

Tabular Q-learning receives three hysteretic bins for each sensor. DQN receives the four-component continuous observation after standardizing the sensor components with statistics collected from 20 random episodes. The tabular controller uses 100,000 interaction steps, $\gamma = 0.98$, and an ϵ-greedy schedule from 1.0 to 0.01. The selected DQN uses 200,000 interaction steps, learning rate 10^{-4}, $\gamma = 0.995$, a replay buffer of 100,000 transitions, batch size 128, target-network update frequency 2,000, and an exploration fraction of 0.3. Shared runtime hyperparameters are defined only in `config.py`.

3.3 Dynamic aerodynamic regime

The dynamic regime tests the same local energy-harvesting question after removing the quasi-steady lookup approximation. Its state comprises three- dimensional position, ground velocity, angle of attack, bank angle, and a continuous wind-frame coordinate. At each 0.1 s integration substep, the environment obtains temporally interpolated wind, forms air-relative velocity, and computes lift, drag, and gravity from the same polar used in the discrete regime. A fourth-order Runge–Kutta integrator advances the coupled flight and wind-frame state. This design preserves the DNS input and wind-scaling rule while exposing the controller to acceleration, transient airspeed, and time-varying aerodynamics absent from the quasi-steady model.

The dynamic observation is

$$o_t = [H_E, \dot{H}_E^{\mathrm{LP}}, \Delta w_n, \mu], \tag{2}$$

where $H_E = z + V_{\mathrm{TAS}}^2/(2g)$ is total-energy height, $\dot{H}_E^{\mathrm{LP}}$ is a one-second low-pass estimate of its rate of change, Δw_n is the wingtip wind difference projected onto the lift normal, and μ is bank angle. The reward is the one-step increment in H_E. This choice makes the training signal and primary energy outcome directly aligned while retaining altitude change as a complementary, more interpretable measure.

The native action has two bounded commands: angle of attack and bank. Both commands pass through first-order actuators with separate time constants and rate limits before affecting the vehicle. PPO receives this continuous action interface through a squashed Gaussian policy. Dynamic DQN instead operates on a 3×3 grid of angle-of-attack and bank commands. The two learned policies therefore share dynamics, observations, reward, initial-scenario sampling, and wind data, but not the same action parameterization. Their comparison is evidence about controller variants in this environment, not an action-space-controlled claim that one RL algorithm is intrinsically better.

Episodes terminate on lower or upper altitude boundaries, numerical divergence, or exhaustion of the DNS sequence. A low true airspeed disables the aerodynamic-force calculation for that integration substep but does not end the episode; numerical divergence is explicitly recorded. The final dynamic analysis will report each termination category. In particular, termination at the upper altitude boundary indicates that the simulated episode ended after energy gain, but does not establish sustained soaring or a safety property.

4 Experimental protocol

The steady study compares random control, tabular Q-learning, and DQN on 100 matched scenarios. A scenario fixes the start frame, initial position, altitude, and heading; all policies in that row receive the same scenario. Scenario generation uses evaluation seed 20260816, and the random-policy action stream is separately seeded. The reported outcome is final minus initial height in metres. We report the mean, median, sample standard deviation, and percentile bootstrap 95% confidence interval for the mean using 20,000 resamples. Matched-scenario differences are summarized with the same bootstrap procedure.

The released steady artifacts contain one selected training run per trainable policy under training seed 1. Consequently, uncertainty intervals describe the finite matched-scenario sample, not variation over

Table 1: Original steady-regime evaluation on 100 matched scenarios. Height change is final minus initial altitude in metres. Intervals are percentile bootstrap 95% confidence intervals for the mean.

Policy	Mean [95% CI]	Median	SD	n
Random	-107.1 $[-162.8, -48.7]$	-186.2	292.8	100
Tabular Q	-5.1 $[-72.0, 64.0]$	-148.4	348.7	100
DQN	42.4 $[-24.8, 111.4]$	49.2	349.4	100

training runs. We reserve multi-seed retraining and a separately generated test-scenario suite as a mandatory follow-up before making confirmatory comparative claims.

4.1 DQN-informed sensor bins

After training the selected DQN, we sampled its action map on a standardized 21×21 grid over $(a_w, \Delta w)$ and fit two thresholds per sensor to its preferred action components. The resulting raw-sensor boundaries are $[-0.8, 0.25]$ for a_w and $[-0.15, 0.60]$ for Δw; the action-map fit error was 0.088. These thresholds replace the original tabular bins, while the tabular training budget and evaluation protocol remain unchanged. This is a representation-design experiment, not policy distillation: the tabular agent is trained independently from a fresh Q table.

The procedure is exploratory because the tuned DQN and the derived bins were selected using the same 100 scenarios subsequently summarized below. The derived-bin result is included to document the hypothesis and to guide the next experiment, not as an out-of-sample comparison.

4.2 Dynamic-regime setup

Dynamic PPO and DQN use the four-dimensional observation above, with each dimension standardized by random-rollout statistics stored with the resulting model artifact. PPO is trained for 100,000 transitions in a 64-environment batch, using 16-step rollouts, GAE, clipped policy updates, and an entropy coefficient annealed from 0.01 to zero. Dynamic DQN is trained for 150,000 transitions with a 100,000-transition replay buffer, batch size 64, Double-DQN targets, Huber loss, and a 3-by-3 action grid. The matched evaluation includes random-grid and fixed-command cruise controls as baselines, together with PPO and DQN. For every scenario, we retain height change, total-energy-height change, episode length, and termination reason.

The existing 100-scenario dynamic result is an exploratory checkpoint. The completion experiment trained three independent replicas of each policy with seeds 11, 22, and 33, then evaluated all frozen replicas on a new scenario suite generated with evaluation seed 20260821. This separates model selection from the paper's comparative estimates. Aggregation treats training seed as the independent replication unit and reports scenario distributions within each seed; it does not pool all episodes across seeds as though they were independent training runs.

5 Results: quasi-steady discrete regime

The original matched-scenario comparison shows a modest ordering in favor of the continuous-observation DQN. Its mean height change was 42.4 m, whereas random control lost 107.1 m on average (Table 1). The paired DQN-minus-random difference was 149.6 m (bootstrap 95% CI $[74.2, 224.4]$). The corresponding tabular-Q-minus-random difference was 102.0 m (95% CI $[25.7, 178.3]$). The DQN mean confidence interval still crosses zero, and all three policies show substantial scenario-to-scenario variation. This baseline therefore supports an energy-harvesting signal in the matched sample but does not establish reliable performance across training seeds.

Hyperparameter tuning substantially improved the selected DQN on the same scenario set, yielding a mean height change of 337.4 m (95% CI $[297.4, 376.2]$). The DQN-informed discretization also improved the tabular result to 192.3 m (95% CI $[130.0, 252.5]$; Table 2). The median of the derived-bin tabular controller, 280.5 m, makes the improvement visible beyond the mean. However, these numbers must not be read as a confirmatory superiority result. The process selected both the DQN candidate and the sensor thresholds using this evaluation suite, and it used a single training seed. The

Table 2: Exploratory steady-regime results on the same 100 scenarios used during selection. These results are not held-out evidence.

Policy	Mean [95% CI]	Median	SD	n
Tuned DQN	337.4 $[297.4, 376.2]$	352.1	201.0	100
DQN-informed tabular Q	192.3 $[130.0, 252.5]$	280.5	315.5	100

Table 3: Held-out dynamic evaluation on 100 matched scenarios. Each learned row is one independent training seed; intervals are percentile bootstrap 95% CIs for the within-seed scenario mean. Height and total-energy-height changes are in metres.

Policy	Seed	Height mean [95% CI]	Energy mean	Median	SD
Random grid	–	-125.1 $[-186.3, -59.9]$	-117.6	-222.3	322.7
Cruise	–	-167.2 $[-216.6, -114.0]$	-165.6	-238.5	261.1
DQN	11	-146.3 $[-197.4, -92.6]$	-131.1	-208.3	265.9
DQN	22	-121.6 $[-173.1, -67.9]$	-104.3	-198.4	272.1
DQN	33	-76.0 $[-132.5, -17.3]$	-59.3	-173.7	297.7
PPO	11	397.7 $[351.3, 440.2]$	396.5	442.8	229.1
PPO	22	276.8 $[212.9, 338.6]$	278.1	399.7	326.7
PPO	33	378.5 $[335.4, 418.7]$	378.6	417.8	212.8

appropriate interpretation is that DQN action boundaries are a promising hypothesis for designing a compact tabular state space.

6 Results: dynamic aerodynamic regime

The dynamic evaluation used 100 matched scenarios generated with evaluation seed 20260821, which was not used to select a model or tune a hyperparameter. Each learned controller was trained independently with seeds 11, 22, and 33; all six frozen artifacts were then evaluated sequentially against the same scenario rows. The replication unit is the training seed, while the 100 scenarios quantify within-replica variation. Table 3 reports the mean, median, sample standard deviation, and scenario-bootstrap interval for each replica. Figure 1 shows the full matched-scenario distributions.

PPO produced positive mean height and total-energy-height changes for all three training seeds. Its mean height changes were 397.7, 276.8, and 378.5 m, with a mean of seed means of 351.0 m and a between-seed standard deviation of 65.0 m. The corresponding total-energy-height changes were 396.5, 278.1, and 378.6 m. PPO therefore preserved energy gain after replacing the quasi-steady lookup with integrated flight and actuator dynamics in this held-out suite.

Dynamic DQN was less consistent in absolute energy gain. Its three mean height changes were -146.3, -121.6, and -76.0 m, and its mean of seed means was -114.6 m (between-seed SD 35.7 m). Its mean total-energy-height change was -98.2 m (between-seed SD 36.3 m). Its three-seed average was less negative than the Random-grid baseline (-125.1 m height, -117.6 m total-energy height) and the fixed Cruise baseline (-167.2 m height, -165.6 m total-energy height), but did not yield net positive energy-height change. Because PPO uses a continuous two-command action and DQN uses a 3×3 discrete grid, this is a comparison of controller variants with different action interfaces, not a pure algorithm ranking.

Termination patterns provide important context. PPO reached the upper altitude boundary in 45–64% of scenarios across seeds, with wind-end fractions of 31–46%. DQN reached the lower altitude boundary in 50–65% of scenarios and the upper boundary in only 4–5%; numerical-divergence fractions were at most 1%. Random grid and Cruise reached the lower boundary in 66% and 73% of scenarios, respectively. An upper-bound termination records that an episode ended after gaining energy in the finite domain; it is not evidence of sustained soaring, safety, or indefinite endurance.

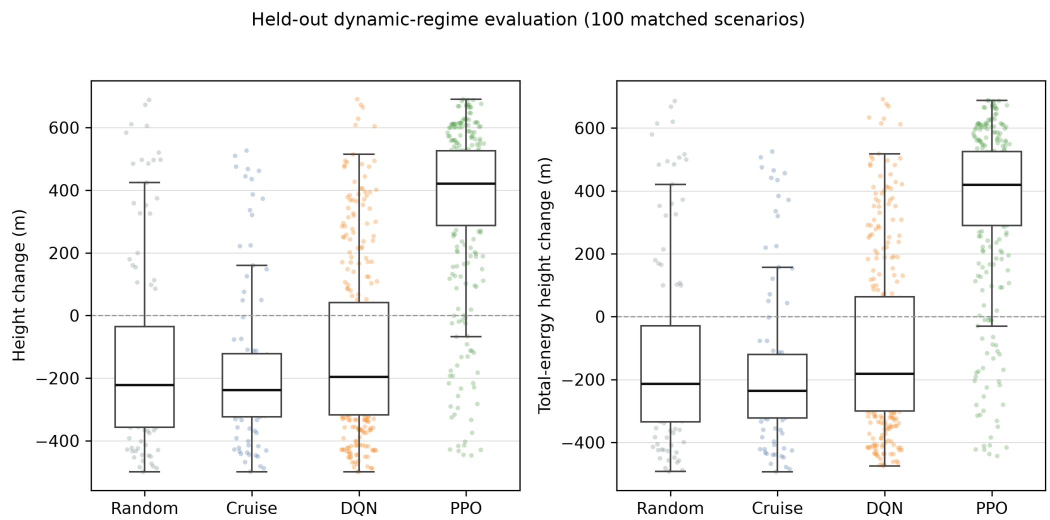

Figure 1: Matched-scenario distributions for the held-out dynamic evaluation. Points retain scenario-level variation; learned-policy points include all three training-seed replicas.

7 Discussion and limitations

The two fidelity regimes form a progression rather than a cross-regime leaderboard. The quasi-steady environment isolates local energy-harvesting decisions with transparent discrete dynamics and was useful for exposing a DQN-informed tabular representation. The dynamic environment then tests the same DNS-driven task with continuous-time flight integration, temporal wind interpolation, actuator lag, and a reward defined directly by total-energy height. The held-out dynamic results show that PPO's positive energy-height signal survives this increase in model fidelity across all three training seeds. Dynamic DQN remained negative in absolute energy-height change, despite a less-negative three-seed average than the non-learning baselines, suggesting that the coarse action grid or value-learning configuration is not yet sufficient for the richer control problem. Since the action interfaces differ, this observation should not be interpreted as an intrinsic PPO-versus-DQN result.

Several limitations bound the claim. The DNS flow is a controlled numerical input at $Ra = 5 \times 10^7$, scaled to the simulated vehicle; it is not an atmospheric measurement. The glider remains a point-mass model driven by a fixed aerodynamic polar, and the dynamic observation contains only local quantities rather than the full flow field. We used three training seeds and one held-out 100-scenario suite; the bootstrap intervals quantify scenario sampling, whereas the small between-seed sample limits inference about training variation. Upper-altitude termination is common for PPO and truncation at the end of the finite wind sequence is also common, so the results demonstrate energy gain over finite episodes rather than sustained safe soaring. No route objective, collision model, deployment constraint, real-flight experiment, or sim-to-real transfer is evaluated.

8 Conclusion

ThermalHunter provides a two-fidelity, DNS-driven platform for studying energy-harvesting soaring in time-varying turbulent wind. In the completed held-out dynamic experiment, PPO replicas achieved positive height and total-energy-height changes across all three training seeds, while dynamic DQN improved over Random-grid and Cruise controls without achieving positive mean energy gain. The results support the narrower claim that reinforcement learning can discover energy-harvesting behavior in this controlled numerical environment. They do not establish route navigation, safety, deployment readiness, sustained soaring, or transfer beyond the simulated flow. Future work should expand training-seed replication, vary the held-out DNS scenarios, and test action representations and dynamic models under explicitly matched control interfaces.

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